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Chemical Equilibrium

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MCQ (Single Correct Answer)	48 questions
Answer key	at the end

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MCQ (Single Correct Answer) 48 questions

Q1. NEET 2026

Given below are certain reactions. Identify the reaction for which $K_P \neq K_C$.

(A) $\mathrm{H_2O(g) + CO(g) \rightleftharpoons H_2(g) + CO_2(g)}$

(B) $\mathrm{N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)}$

(C) $\mathrm{H_2(g) + I_2(g) \rightleftharpoons 2HI(g)}$

(D) $\mathrm{N_2(g) + O_2(g) \rightleftharpoons 2NO(g)}$

Open solution

Q2. NEET 2025

For the reaction $\mathrm{A(g) \rightleftharpoons 2\,B(g)}$, the backward reaction rate constant is higher than the forward reaction rate constant by a factor of 2500 , at 1000 K.

[Given : $\mathrm{R = 0.0831\ L\ atm\ mol^{-1}\ K^{-1}}$]

K_p for the reaction at $1000K$ is

(A) 0.033 (B) 0.021 (C) 83.1 (D) 2.077×10^5

Open solution

Q3. NEET 2025

Higher yield of NO in $\mathrm{N_2(g) + O_2(g) \rightleftharpoons 2NO(g)}$ can be obtained at $\left[\Delta \mathrm{H}\text{ of the reaction } = +180.7\ \mathrm{kJ\ mol^{-1}}\right]$

A. Higher temperature

B. Lower temperature

C. Higher concentration of $\mathrm{N_2}$

D. Higher concentration of $\mathrm{O_2}$

Choose the correct answer from the options given below :

(A) B, C, D only (B) A, C, D only (C) A, D only (D) B, C only

Open solution

Q4. NEET 2024 (Re-Examination)

At a given temperature and pressure, the equilibrium constant values for the equilibria are given below:

$\mathrm{3\,A_2 + B_2 \rightleftharpoons 2\,A_3\,B,\ K_1}$

$\mathrm{A_3\,B \rightleftharpoons \frac{3}{2}\,A_2 + \frac{1}{2}\,B_2,\ K_2}$

The relation between $\mathrm{K_1}$ and $\mathrm{K_2}$ is :

(A) $\mathrm{K_1^2 = 2\,K_2}$ (B) $\mathrm{K_2 = \frac{K_1}{2}}$

(C) $\mathrm{K_1 = \frac{1}{\sqrt{K_2}}}$ (D) $\mathrm{K_2 = \frac{1}{\sqrt{K_1}}}$

Open solution

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Q5. NEET 2024 (Re-Examination)

For the reaction in equilibrium

$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g), \Delta H = -Q$

Reaction is favoured in forward direction by:

(A) use of catalyst

(B) decreasing concentration of N_2

(C) low pressure, high temperature and high concentration of ammonia

(D) high pressure, low temperature and higher concentration of H_2

Open solution

Q6. NEET 2024

In which of the following equilibria, K_p and K_c are NOT equal?

(A) $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$

(B) $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$

(C) $CO_{(g)} + H_2O_{(g)} \rightleftharpoons CO_{2(g)} + H_{2(g)}$

(D) $2BrCl_{(g)} \rightleftharpoons Br_{2(g)} + Cl_{2(g)}$

Open solution

Q7. NEET 2024

For the reaction $2A \rightleftharpoons B + C, K_c = 4 \times 10^{-3}$. At a given time, the composition of reaction mixture is: $[A] = [B] = [C] = 2 \times 10^{-3}M$. Then, which of the following is correct?

(A) Reaction is at equilibrium.

(B) Reaction has a tendency to go in forward direction.

(C) Reaction has a tendency to go in backward direction.

(D) Reaction has gone to completion in forward direction.

Open solution

Q8. NEET 2024

Consider the following reaction in a sealed vessel at equilibrium with concentrations of $N_2 = 3.0 \times 10^{-3}M, O_2 = 4.2 \times 10^{-3}M$ and $NO = 2.8 \times 10^{-3}M$.

$2NO_{(g)} \rightleftharpoons N_{2(g)} + O_{2(g)}$

If $0.1\ mol\ L^{-1}$ of $NO_{(g)}$ is taken in a closed vessel, what will be degree of dissociation (α) of $NO_{(g)}$ at equilibrium?

(A) 0.00889 (B) 0.0889 (C) 0.8889 (D) 0.717

Open solution

Q9. NEET 2023 Manipur

For a weak acid HA, the percentage of dissociation is nearly 1% at equilibrium. If the concentration of acid is 0.1 mol L^{-1}, then the correct option for its K_a at the same temperature is :

(A) 1×10^{-4} (B) 1×10^{-6} (C) 1×10^{-5} (D) 1×10^{-3}

Open solution

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Q10. NEET 2022 Phase 2

K_p for the following reaction is 3.0 at 1000 K.

$CO_2(g) + C(s) \rightleftharpoons 2CO(g)$

What will be the value of K_c for the reaction at the same temperature?

(Given : R = 0.083 L bar K^{-1} mol^{-1})

(A) 3.6 (B) 0.36 (C) 3.6×10^{-2} (D) 3.6×10^{-3}

Open solution

Q11. NEET 2022 Phase 1

$3O_2(g) \rightleftharpoons 2O_3(g)$

for the above reaction at 298 K, K_c is found to be 3.0×10^{-59}. If the concentration of O_2 at equilibrium is 0.040 M then concentration of O_3 in M is

(A) 4.38×10^{-32} (B) 1.9×10^{-63} (C) 2.4×10^{31} (D) 1.2×10^{21}

Open solution

Q12. NEET 2018

Which one of the following conditions will favour maximum formation of the product in the reaction $A_{2(g)} + B_{2(g)} \rightleftharpoons X_{2(g)}$, $\Delta_r H = -X$ kJ ?

(A) Low temperature and high pressure
(B) Low temperature and low pressure
(C) High temperature and high pressure
(D) High temperature and low pressure

Open solution

Q13. NEET 2017

The equilibrium constants of the following are

$N_2 + 3H_2 \rightleftharpoons 2NH_3$; K_1

$N_2 + O_2 \rightleftharpoons 2NO$; K_2

$H_2 + \frac{1}{2}O_2 \rightleftharpoons H_2O$; K_3

The equilibrium constant (K) of the reaction :

$2NH_3 + \frac{5}{2} O_2 \rightleftharpoons 2NO + 3H_2O$ will be

(A) $K_2K_3^3/K_1$ (B) K_2K_3/K_1 (C) $K_2^3K_3/K_1$ (D) $K_1K_3^3/K_2$

Open solution

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Q14. NEET 2017

A 20 litre container at 400 K contains $CO_{2(g)}$ at pressure 0.4 atm and an excess of SrO (neglect the volume of solid SrO). The volume of the container is now decreased by moving the movable piston fitted in the container. The maximum volume of the container, when pressure of CO_2 attains its maximum value, will be

(Given that : $SrCO_{3(s)} \rightleftharpoons SrO_{(s)} + CO_{2(g)}$, K_p = 1.6 atm)

(A) 10 litre (B) 4 litre (C) 2 litre (D) 5 litre

Open solution

Q15. NEET 2016 Phase 1

Consider the following liquid-vapour equilibrium.
Liquid $\rightleftharpoons$ Vapour
Which of the following relations is correct ?

(A) $\frac{d\ln P}{dT^2} = \frac{-\Delta H_v}{T^2}$ (B) $\frac{d\ln P}{dT} = \frac{\Delta H_v}{RT^2}$
(C) $\frac{d\ln G}{dT^2} = \frac{\Delta H_v}{RT^2}$ (D) $\frac{d\ln P}{dT} = \frac{-\Delta H_v}{RT}$

Open solution

Q16. AIPMT 2015

If the equilibrium constant for

$N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$ is K, the equilibrium

constant for

$\frac{1}{2} N_{2(g)} + \frac{1}{2} O_{2(g)} \rightleftharpoons NO_{(g)}$ will be

(A) $\frac{1}{2}$ K (B) K (C) K^2 (D) $K^{1/2}$

Open solution

Q17. AIPMT 2015 Cancelled Paper

Which of the following statements is correct for a reversible process in a state of equilibrium?

(A) $\Delta G^o = -2.30$ RT log K (B) $\Delta G^o = 2.30$ RT log K
(C) $\Delta G = -2.30$ RT log K (D) $\Delta G = 2.30$ RT log K

Open solution

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Q18. AIPMT 2015 Cancelled Paper

If the value of equilibrium constant for a particular reaction is 1.6×10^{12}, then at equilibrium the system will contain

(A) mostly products

(B) similar amounts of reactants and products

(C) all reactants

(D) mostly reactants.

Open solution

Q19. AIPMT 2014

Using the Gibb's energy change, ΔG^o = +63.3 kJ, for the following reaction,

$Ag_2CO_{3(s)} \rightleftharpoons 2\ Ag^+_{(aq)} + CO_3{}^{2-}{}_{(aq)}$

the K_{sp} of $Ag_2CO_{3(s)}$ in water at 25°C is

($R = 8.314\ J\ K^{-1}\ mol^{-1}$)

(A) 3.2×10^{-26} (B) 8.0×10^{-12} (C) 2.9×10^{-3} (D) 7.9×10^{-2}

Open solution

Q20. AIPMT 2014

For the reversible reaction,

$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$ + heat

The equilibrium shifts in forward direction

(A) by increasing the concentration of $NH_{3(g)}$

(B) by decreasing the pressure

(C) by decreasing the concentrations of $N_{2(g)}$ and $H_{2(g)}$

(D) by increasing pressure and decreasiing temperature.

Open solution

Q21. AIPMT 2014

For a given exothermic reaction, K_p and K'_p are the equilibrium constants at temperatures T_1 and T_2, respectively. Assuming that heat of reaction is constant in temperature range between T_1 and T_2, it is readily observed that

(A) $K_p > K'_p$ (B) $K_p < K'_p$ (C) $K_p = K'_p$ (D) $K_p = \frac{1}{k'_p}$

Open solution

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Q22. AIPMT 2012 Mains

Open solution

Given that the equilibrium constant for the reaction,
$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$
has a value of 278 at a particular temperature. What is the value of the equilibrium constant for the following reaction at the same temperature ?
$SO_{3(g)} \rightleftharpoons SO_{2(g)} + \frac{1}{2} O_{2(g)}$

(A) 1.8×10^{-3} (B) 3.6×10^{-3} (C) 6.0×10^{-2} (D) 1.3×10^{-5}

Q23. AIPMT 2012 Mains

Open solution

Given the reaction between 2 gases represented by A_2 and B_2 to give the compound $AB_{(g)}$,

$A_2(g) + B_{2(g)} \rightleftharpoons 2AB_{(g)}$

At equilibrium, the concentration of
$A_2 = 3.0 \times 10^{-3}$ M, of $B_2 = 4.2 \times 10^{-3}$ M, of AB = 2.8×10^{-3} M
If the reaction takes place in a sealed vessel at 527°C, then the value of K_c will be

(A) 2.0 (B) 1.9 (C) 0.62 (D) 4.5

Q24. AIPMT 2011 Prelims

Open solution

The value of ΔH for the reaction
$X_{2(g)} + 4Y_{2(g)} \rightleftharpoons 2XY_{4(g)}$
is less than zero. Formation of $XY_{4(g)}$ will be favoured at

(A) high temperature and high pressure (B) low pressure and low temperature
(C) high temperature and low pressure (D) high pressure and low temperature

Q25. AIPMT 2011 Prelims

Open solution

For the reaction, $N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$, the equilibrium constant is K_1. The equilibrium constant is K_2 for the reaction,
$2NO_{(g)} + O_{2(g)} \rightleftharpoons 2NO_{2(g)}$
What is K for the reaction,
$NO_{2(g)} \rightleftharpoons \frac{1}{2} N_{2(g)} + O_{2(g)}$

(A) $\frac{1}{2K_1K_2}$ (B) $\frac{1}{4K_1K_2}$
(C) $\left[\frac{1}{K_1K_2}\right]^{1/2}$ (D) $\frac{1}{K_1K_2}$

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Q26. AIPMT 2010 Mains

Open solution

The reaction,
$2A_{(g)} + B_{(g)} \rightleftharpoons 3C_{(g)} + D_{(g)}$
is begun with the concentrations of A and B both at an initial value of 1.00 M. When equilibrium is reached, the concentration of D is measuread and found to be 0.25 M. The value for the equilibrium constant for this reaction is given by the expression

(A) $[(0.75)^3\ (0.25)] \div [(1.00)^2\ (1.00)]$
(B) $[(0.75)^3\ (0.25)] \div [(0.50)^2\ (0.75)]$
(C) $[(0.75)^3\ (0.25)] \div [(0.50)^2\ (0.25)]$
(D) $[(0.75)^3\ (0.25)] \div [(0.75)^2\ (0.25)]$

Q27. AIPMT 2010 Prelims

Open solution

In which of the following equilibrium K_c and K_p are not equal?

(A) $2NO_{(g)} \rightleftharpoons N_{2(g)} + O_{2(g)}$
(B) $SO_{2(g)} + NO_{2(g)} \rightleftharpoons SO_{3(g)} + NO_{(g)}$
(C) $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$
(D) $2C_{(s)} + O_{2(g)} \rightleftharpoons 2CO_{2(g)}$

Q28. AIPMT 2009

Open solution

The dissociation constants for acetic acid and HCN at 25°C are 1.5×10^{-5} and 4.5×10^{-10} respectively. The equilibrium constant for the equilibrium
$CN^- + CH_3COOH \rightleftharpoons HCN + CH_3COO^-$ would be

(A) 3.0×10^{-5}
(B) 3.0×10^{-4}
(C) 3.0×10^{4}
(D) 3.0×10^{5}

Q29. AIPMT 2008

Open solution

The dissociation equilibrium of a gas AB_2 can be represented as :
$2AB_{2(g)} \rightleftharpoons 2AB_{(g)} + B_{2(g)}$
The degree of dissociation is x and is small compared to 1. The expression relating the degree of dissociation (x) with equilibrium constant K_p and total pressure P is

(A) $(2K_p/P)^{1/2}$
(B) (K_p/P)
(C) $(2K_p/P)$
(D) $(2K_p/P)^{1/3}$

Q30. AIPMT 2008

Open solution

If the concentration of OH^- ions in the reaction
$Fe(OH)_{3(s)} \rightleftharpoons Fe^{3+}_{(aq)} + 3OH^-_{(aq)}$
is decreased by 1/4 times, then equilibrium concentration of Fe^{3+} will increase by

(A) 64 times
(B) 4 times
(C) 8 times
(D) 16 times

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Q31. AIPMT 2008

The value of equilibrium constant of the reaction
$HI_{(g)} \rightleftharpoons \frac{1}{2}H_{2(g)} + \frac{1}{2}I_{2(g)}$
is 8.0. The The equilibrium constant of the reaction
$H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$ will be

(A) 16 (B) 1/8 (C) 1/16 (D) 1/64

Open solution

Q32. AIPMT 2008

The values of K_p for the reactions,

$X \rightleftharpoons Y + Z$ . . . .(i)
$A \rightleftharpoons 2B$ . . . .(ii)

are in the ratio 9 : 1. If degree of dissociation of X and A be equal, then total pressure at equilibrium (i) and (ii) are in the ratio

(A) 36 : 1 (B) 1 : 1 (C) 3 : 1 (D) 1 : 9

Open solution

Q33. AIPMT 2007

The equilibrium constants of the following are

$N_2 + 3H_2 \rightleftharpoons 2NH_3$; K_1

$N_2 + O_2 \rightleftharpoons 2NO$; K_2

$H_2 + \frac{1}{2}O_2 \rightleftharpoons H_2O$; K_3

The equilibrium constant (K) of the reaction :

$2NH_3 + \frac{5}{2}O_2 \rightleftharpoons 2NO + 3H_2O$ will be

(A) $K_2K_3^3/K_1$ (B) K_2K_3/K_1 (C) $K_2^3K_3/K_1$ (D) $K_1K_3^3/K_2$

Open solution

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Q34. AIPMT 2006

Open solution

For the reaction :

$CH4_{(g)} + 2O_{2(g)} \rightleftharpoons CO_{2(g)} + 2H_2O_{(l)}$,

$\Delta H_r = -170.8$ kJ mol^{-1}.

Which of the following statements is not true?

(A) The reaction is exothermic.

(B) At equilibrium, the concentrations of $CO_{2(g)}$ and $H_2O_{(l)}$ are not equal.

(C) The equilibrium constant for the reaction is given by $K_p = \frac{[CO_2]}{[CH_4][O_2]}$

(D) Addition of $CH_{4(g)}$ or $O_{2(g)}$ at equilibrium will cause a shift to the right.

Q35. AIPMT 2005

Open solution

Equilibrium constants K_1 and K_2 for the following equilibria:

$$NO_{(g)} + \frac{1}{2}O_2 \overset{K_1}{\rightleftharpoons} NO_{2(g)} \text{ and}$$

$$2NO_{2(g)} \overset{K_2}{\rightleftharpoons} 2NO_{(g)} + O_{2(g)}$$

are related as

(A) $K_2 = 1/K_1^2$ (B) $K_2 = K_1^2$ (C) $K_2 = 1/K_1$ (D) $K_2 = K_1/2$

Q36. AIPMT 2003

Open solution

The equilibrium constants of the following are

$N_2 + 3H_2 \rightleftharpoons 2NH_3$; K_1

$N_2 + O_2 \rightleftharpoons 2NO$; K_2

$H_2 + \frac{1}{2}O_2 \rightleftharpoons H_2O$; K_3

The equilibrium constant (K) of the reaction :

$2NH_3 + \frac{5}{2}O_2 \rightleftharpoons 2NO + 3H_2O$ will be

(A) $K_2K_3^3/K_1$ (B) K_2K_3/K_1 (C) $K_2^3K_3/K_1$ (D) $K_1K_3^3/K_2$

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Q37. AIPMT 2003

The reaction quotient (Q) for the reaction

$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$ is given by

$Q = \frac{[NH_3]^2}{[N_2][H_2]^3}$.

The reaction will proceed from right to left if

(A) $Q = K_c$ (B) $Q < K_c$ (C) $Q > K_c$ (D) $Q = 0$

Open solution

Q38. AIPMT 2002

Reaction $BaO_{2(s)} \rightleftharpoons BaO_{(s)} + O_{2(g)}$; ΔH = +ve. In equilibrium condition, pressure of O_2 depends on

(A) increase mass of BaO_2

(B) increase mass of BaO

(C) increase temperature on equilibrium

(D) increase mass of BaO_2 and BaO both.

Open solution

Q39. AIPMT 2000

For any reversible reaction, if we increase concentration of the reactants, then effect on equilibrium constant

(A) depends on amount of concentration

(B) unchanged

(C) decrease

(D) increase

Open solution

Q40. AIPMT 2000

Equilibrium constant K_p for following reaction

$MgCO_{3(s)} \rightleftharpoons MgO_{(s)} + CO_{2(g)}$

(A) $K_p = P_{CO_2}$

(B) $K_p = P_{CO_2} \times \frac{P_{CO_2} \times P_{MgO}}{P_{MgCO_3}}$

(C) $K_p = \frac{P_{CO_2} + P_{MgO}}{P_{MgCO_3}}$

(D) $K_p = \frac{P_{MgCO_3}}{P_{CO_2} + P_{MgO}}$

Open solution

Q41. AIPMT 1998

If K_1 and K_2 are the respective equilibrium constants for the two reactions

$XeF_6(g) + H_2O(g) \rightleftharpoons XeOF_4(g) + 2HF(g)$ $XeO_4(g) + XeF_6(g) \rightleftharpoons XeOF_4(g) + XeO_3F_2(g)$

the equilibrium constant of the reaction

$XeO_4(g) + 2HF(g) \rightleftharpoons XeO_3F_2(g) + H_2O(g)$

will be

(A) $K_1/(K_2)^2$ (B) $K_1 \cdot K_2$ (C) K_1/K_2 (D) K_2/K_1

Open solution

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Q42. AIPMT 1996

The equilibrium constant for the reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$ is K, then the equilibrium constant for the equilibrium $2NH_3 \rightleftharpoons N_2 + 3H_2$ is

(A) $\sqrt{K}$ (B) $\sqrt{\frac{1}{K}}$ (C) $\frac{1}{K}$ (D) $\frac{1}{K^2}$

Open solution

Q43. AIPMT 1996

If α is dissociation constant, then the total number of moles for the reaction, $2HI \rightarrow H_2 + I_2$ will be

(A) 1 (B) $1 - \alpha$ (C) 2 (D) $2 - \alpha$

Open solution

Q44. AIPMT 1995

Standard state Gibbs free energy change for isomerization reaction *cis*-2-pentene $\rightleftharpoons$ *trans*-2-pentene is $-3.67\ kJ/mol$ at 400 K. If more *trans*-2-pentene is added to the reaction vessel, then

(A) equilibrium remains unaffected

(B) equilibrium is shifted in the forward direction

(C) more *cis*-2-pentene is formed

(D) additional *trans*-2-pentene is formed.

Open solution

Q45. AIPMT 1995

In liquid-gas equilibrium, the pressure of vapours above the liquid is constant at

(A) constant temperature (B) low temperature

(C) high temperature (D) none of these.

Open solution

Q46. AIPMT 1993

According to Le Chatelier's principle, adding heat to a solid and liquid in equilibrium will cause the

(A) temperature to increase (B) temperature to decrease

(C) amount of liquid to decrease (D) amount of solid to decrease.

Open solution

Q47. AIPMT 1992

Which one of the following information can be obtained on the basis of Le Chatelier principle?

(A) Dissociation constant of a weak acid

(B) Entropy change in a reaction

(C) Equilibrium constant of a chemical reaction

(D) Shift in equilibrium position on changing value of a constraint

Open solution

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Q48. AIPMT 1989

Open solution

K_1 and K_2 are equilibrium constant for reactions (i) and (ii)

$N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$ ....(i)

$NO_{(g)} \rightleftharpoons \frac{1}{2}N_{2(g)} + \frac{1}{2}O_{2(g)}$ ...(ii)

(A) $K_1 = \left(\frac{1}{K_2}\right)^2$

(B) $K_1 = K_2^2$

(C) $K_1 = \frac{1}{K_2}$

(D) $K_1 = (K_2)^0$

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Answer Key

1. **B**	2. **A**	3. **B**	4. **D**	5. **D**
6. **A**	7. **C**	8. **D**	9. **C**	10. **C**
11. **A**	12. **A**	13. **A**	14. **D**	15. **B**
16. **D**	17. **A**	18. **A**	19. **B**	20. **D**
21. **A**	22. **C**	23. **C**	24. **D**	25. **C**
26. **B**	27. **D**	28. **C**	29. **D**	30. **A**
31. **D**	32. **A**	33. **A**	34. **C**	35. **A**
36. **A**	37. **C**	38. **C**	39. **B**	40. **A**
41. **D**	42. **C**	43. **C**	44. **C**	45. **A**
46. **D**	47. **D**	48. **A**		

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